Intro to Statistical Methods

Assignment - 2

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Section 7

1) Correlation between Exercise duration and Resting Heart Rate

X => Exercise duration (min)

Y → Resting Heart Rate (bpm)

a) Computational Table

S. no	X	Y	X^2	Y^2	XY
1	15	85	225	7225	1275
2	20	82	400	6724	1640
3	25	81	625	6561	2025
4	30	77	900	5929	2310
5	35	76	1225	5776	2660
6	40	74	1600	5476	2960
7	45	73	2025	5329	3285
8	50	71	2500	5041	3550
9	55	69	3025	4761	3795
10	60	67	3600	4489	4020
11	65	65	4225	4225	4225
Σ	440	820	20350	61536	31745

b) $\Sigma X = 440$ $\qquad \bar{X} = \frac{440}{11} = 40$

$\Sigma Y = 820$ $\qquad \bar{Y} = 820/11 = 74.545$

$\Sigma X^2 = 20350$ $\qquad n = 11$

$\Sigma Y^2 = 61536$

$\Sigma XY = 31745$

c) Pearson's coefficient of correlation

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$$r = \frac{\sum xy - \frac{\sum x \sum y}{n}}{\sqrt{\left[\sum x^2 - \frac{(\sum x)^2}{n}\right]\left[\sum y^2 - \frac{(\sum y)^2}{n}\right]}}$$

$$= \frac{31745 - \frac{440 \times 820}{11}}{\sqrt{\left(20350 - \frac{440^2}{11}\right)\left(61536 - \frac{820^2}{11}\right)}}$$

$$= \frac{31745 - 32800}{\sqrt{(20350 - 17600)(61536 - 61127.27)}}$$

$$= \frac{-1055}{1060.19}$$

$$r \approx -0.9951$$

d) $r = -ve$ ∴ inverse relation

as x increases y decreases

$|r|$ is close to 1, very strong linear relationship

$$r^2 = (-0.9951)^2 \approx 0.990$$

coefficient of determination = 0.99

∴ 99% of the variation can be explained by x

e) Based on strong statistical evidence ($r = -0.995$, $r^2 = 0.99$), longer exercise duration is associated with lower resting heart rate, a recognised marker of better cardio-vascular fitness. It is therefore reasonable to recommend that the wellness cordinator encourage participants to increase their regular daily exercise duration.

2) One way ANOVA on Branch waiting times:

a)

Customer	Branch A	Branch B	Branch C
1	18	15	12
2	20	16	13
3	19	17	14
4	21	18	15
5	22	16	13
6	20	17	14
	ΣX 120	99	81

$N_{tot} = n_A + n_B + n_C = 18$

$\Sigma X_A = 120$ $\Sigma X_B = 99$ $\Sigma X_C = 81$

$\Sigma X_{tot} = 300$

$\bar{X}_A = 120/6 = 20$

$\bar{X}_B = 99/6 = 16.50$

$\bar{X}_C = 81/6 = 13.5$

$\bar{X}_{tot} = 300/18 = 16.67$

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b) $CF = \dfrac{\sum X_{tot}^2}{n_{tot}}$

$= \dfrac{300^2}{18} = 5000$

$\sum\sum x^2 = \sum X_a^2 + \sum X_b^2 + \sum X_c^2$

$= (18^2 + 20^2 + 19^2 + 21^2 + 22^2 + 20^2) + (15^2 + 16^2 + 17^2 + 18^2 + 16^2 + 17^2) + (12^2 + 13^2 + 14^2 + 15^2 + 13^2 + 14^2)$

$= 2410 + 1639 + 1099$

$= 5148$

$SST = \sum\sum x^2 - CF$

$= 5148 - 5000$

$= 148$

$SSB = \sum\left(\dfrac{(\sum x_i)^2}{n_i}\right) - CF$

$= \left(\dfrac{120^2}{6} + \dfrac{99^2}{6} + \dfrac{81^2}{6}\right) - 5000$

$= 5127 - 5000$

$= 127$

$SSW = SST - SSB$

$= 148 - 127$

$= 21$

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c) $SSB = 127$

$df_b = k - 1$

$= 2$

~~Mean~~ $MS_b = \frac{SSB}{df_b}$

$= 127/2 = 63.5$

$SSW = 21$

$df_w = N - k$

$= 18 - 3$

$= 15$

$MS_w = 21/15 = 1.4$

$F = \frac{MS_b}{MS_w} = \frac{63.5}{1.4} = 45.36$

ANOVA table

Source of Var	SS	df	MS	F
Between branches	127	2	63.5	45.36
within branches	21	15	1.4	
Tot	148	17		

d) $H_0 : \mu_A = \mu_B = \mu_C$

H_1 : atleast one branch mean differs from others

$\alpha = 0.05$

$df_b = 2 \quad df_w = 15$

$F_{0.05}(2, 15) = 3.68$ (F table)

$F_{calculated} = 45.36$

$F_{critical} = 3.68$

Since $F_{calculated} > F_{critical}$

reject H_0

∴ A 5% level of significance, there is sufficient statistical evidence to conclude that the average customer waiting time differs significantly between branches

e) The ANOVA results show a highly significant difference in mean waiting times

$\bar{X}_C = 13.5$ (lowest)

$\bar{X}_B = 16.5$

$\bar{X}_A = 20$ (highest)

Branch C recorded the lowest average waiting time, suggesting that the operational strategy in place there is currently the most effective at reducing customer waiting times.

Management should consider rolling out the Branch C's specific practices to other 2 branches.